

Process Creation Investigation – sysmon.exe

Objective

The objective of this investigation is to verify that sysmon.exe process execution events are successfully collected by Elastic SIEM using Sysmon logs. This investigation demonstrates how Sysmon activity can be monitored to validate endpoint visibility and detect potentially suspicious system monitoring or administrative actions.

Command Executed

sysmon

Detection Method (KQL)

`process.name : "sysmon.exe"`

If your environment indexes the event differently, use the query that successfully located the event.

Evidence Collected

Timestamp: 2026-07-14T06:21:03.367Z

Process Name: sysmon.exe

Host Name: sepisecurity

User Domain: Sepisecurity

Process ID (PID): 37456

Event Code: 1 (Sysmon Process Creation)

Analysis

The investigation confirmed that sysmon.exe executed successfully on the monitored Windows endpoint. The collected evidence identified the executing user, host system, process identifier, and execution time. No suspicious parent process, unusual command-line arguments, or indicators of malicious behavior were observed. The activity is consistent with legitimate execution of the Sysmon utility for endpoint monitoring.

Security Impact

Sysmon is a legitimate Microsoft Sysinternals tool that provides detailed endpoint telemetry for security monitoring. Attackers may attempt to disable, reconfigure, or abuse Sysmon to evade detection. Monitoring sysmon.exe execution helps SOC analysts verify monitoring integrity and identify unusual activity affecting endpoint visibility.

Conclusion

This investigation demonstrates the ability to detect and analyze Windows process creation events using Elastic SIEM and Sysmon. The collected evidence indicates legitimate administrative activity, and no indicators of compromise were identified. This investigation highlights practical skills in KQL searching, endpoint telemetry analysis, and professional SOC documentation.